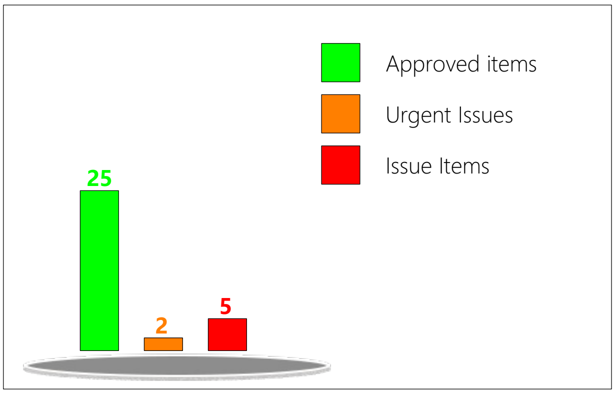
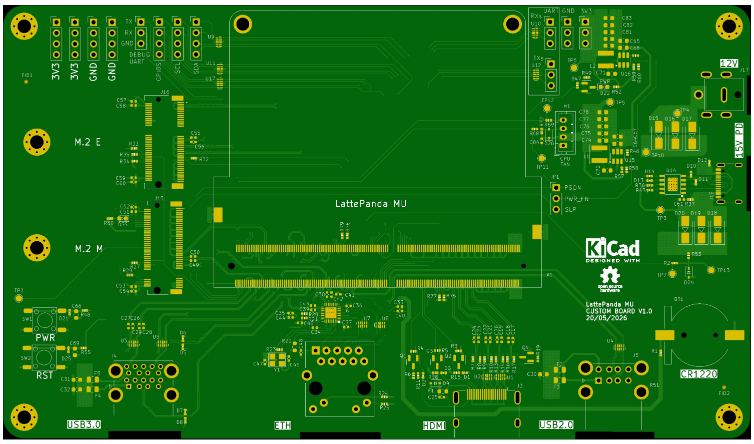
HQDFM Design for Manufacture(DFM) Report

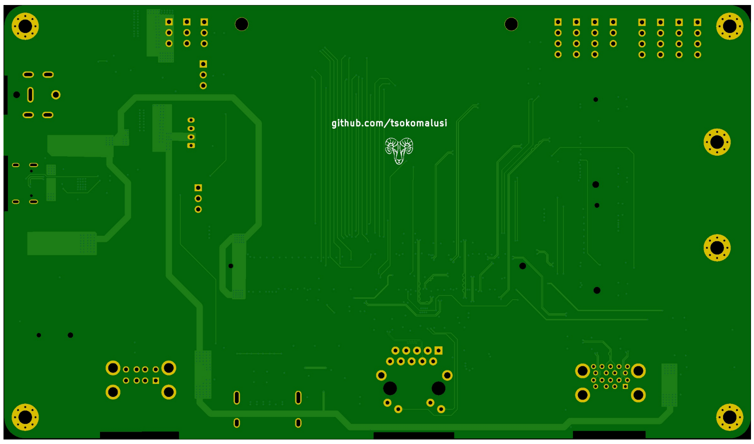
File name: Gerber

Time: 2026-05-25 Layer count:4 PCB Thickness: 1.60 Quantity: 5 mm



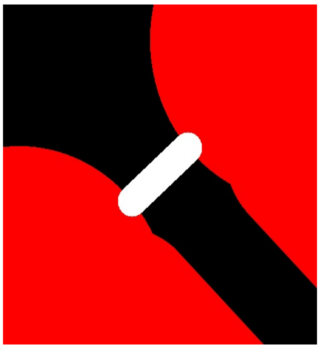
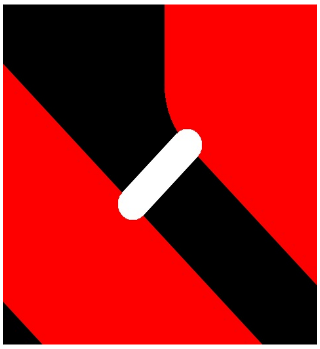
Basic Board Specs	Trace Width/Spacing	6.00/4.00mil
	Milling Density	31.4342m/m²
	Surface Finish Area	12.08%
	Test Point Count	1068

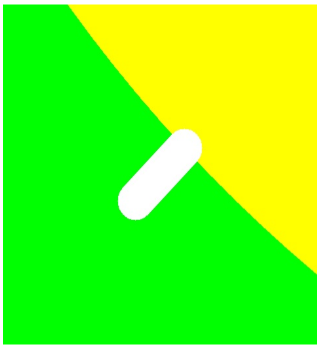
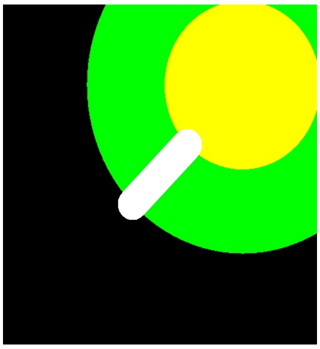


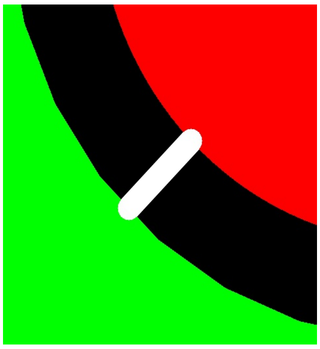
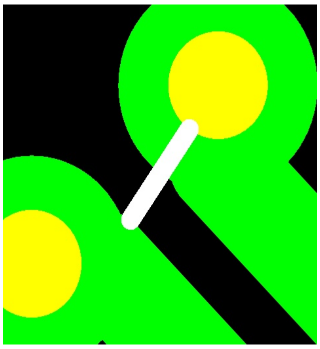


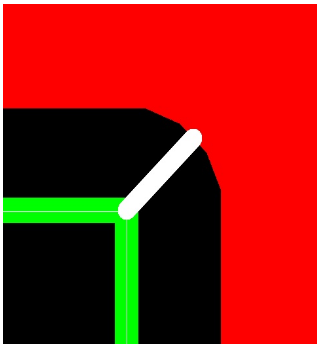
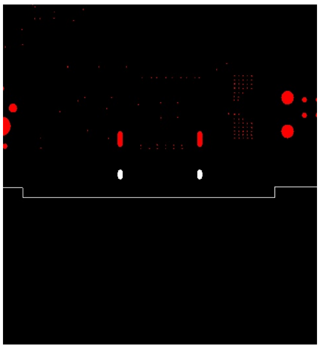
Type	Category	No. of Checks	Result
PCB Trace Analysis	Open/Shorts (IPC)	1	Fail
	Signal Integrity	4	Pass
	Smallest Trace Width	1	Pass 10
	Smallest Trace Spacing	3	Pass 581 , Fail 319
	SMD Pad Spacing	1	Pass
	Pad Size	3	Pass 54
	Hatched Copper Pour	2	Pass
	Annular Ring Size	2	Pass 3 , Fail 3
	Drill to Copper	5	Pass 1473 , Fail 42
	Copper-to-Board Edge	2	Pass 3 , Fail 56
	Holes on SMD Pads	4	Pass
PCB Drilling Analysis	Drill Diameter	8	Pass 61 , Fail 2
	Drill Spacing	4	Pass 457 , Fail 2
	Drill to Board Edge	4	Pass 2
	Drill Hole Density	1	Pass
	Special Drill Holes	2	Pass
	Drill Hole Errors	3	Pass
PCB Solder Mask Analysis	Solder Mask Dam	2	Pass 15
	Missing SMask Opening	1	Pass
	Solder Paste Area	1	Pass
PCB Silk Analysis	Silkscreen Spacing	1	Pass 15

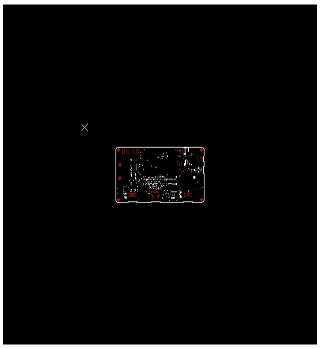
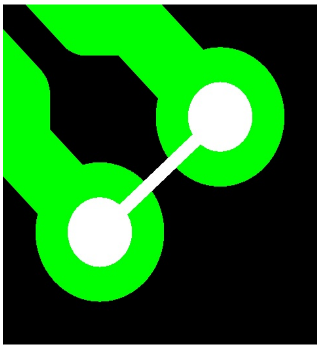
PCBA Component Analysis	Component Spacing	1	Fail
	Comp.-to-Board-Edge	3	Fail
	Component Silkscreen Spacing	0	Fail
	Pad Count Mismatch	4	Fail
	Designator Length	0	Fail
	Double-sided Components	1	Pass
	Mid Mount Comp	1	Pass
	Comp Height Analysis	2	Pass
PCBA Pin Analysis	SMT Pad Analysis	8	Fail
	Through-hole Pins	10	Fail
PCBA Pad Analysis	Chip Pads	60	Fail
PCBA Fiducial Analysis	Fiducial Count	1	Fail
	Fiducial Analysis	4	Pass

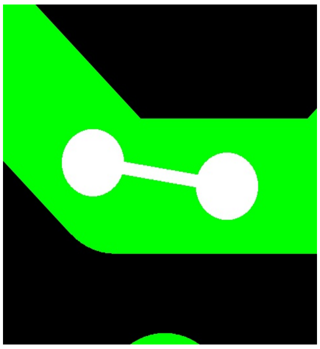
ID	Check	Limits	Value	Issue	Image	Position	Qty	Level
1	Smallest Trace Spacing_SMD Pad Spacing	4,6,10	3.89 mil	For most factories, the minimum pad-to-pad spacing requirement is around 3-4 mil. Failure to meet the factory's requirement could result in incomplete solder mask dams between the pads, thereby increasing the risk of incomplete etching of the pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Pad-to-pad spacings of 3.89mil were detected in your design. The spacing should be at least 6 mil where possible. For solder mask covered pads, 4 mil spacing is sufficient.		123.71,-97.42	7	Risk
2	Smallest Trace Spacing_Trace Spacing	3.5,4,5	0.10 mm	Trace-to-trace spacing of 3.89mil were detected in your design. This could result in incomplete etching between the traces, thereby producing shorts, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 6 mil for regular routing and at least 4 mil in high density areas, such as when routing fine pitch BGAs.		112.73,-69.45	81	Risk

3	Annular Ring Size_PTH Annular Ring	6,7,8	0.07 mm	PTH Pad Rings2.93 mil in size were detected in your design. This could increase the risk of tangency or breakouts, which decrease electrical reliability and yield, and affect the reliability of the boards. It is recommended that PTH Pad Rings have a width of at least 6 mil.		118.16,-42.35	4	Risk
4	Annular Ring Size_Via Annular Ring	4,5,6	0.07 mm	Min via annular rings3.91mil in size were detected in your design.It will affect production efficiency and electrical reliability. It is recommended that the minimum ring size for "via annular rings" be ≥5 mils.		66.61,-122.09	2	Risk

5	Drill to Copper_NPTH-to-Copper	8,10,12	0.15 mm	The NPTH to copper spacing should be at least 8 mil (ideally 12 mil). Spacing less than this could increase the risk of defects such as exposed copper, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The NPTH to copper spacing in your design is only 5.90mil. It is recommended to increase the spacing to at least 12 mil.		98.95,-59.43	36	Risk
6	Drill to Copper_Via-to-Trace [Outer]	8,10,12	0.15 mm	Via-to-trace spacing (outer layer) of 8.07mil was detected in your design. This could increase the risk of defects such as short circuits, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 10 mil.		123.78,-97.42	431	Risk

7	Copper-to-Board Edge_Copper-to-Board Edge	8,11,15	0.20 mm	For most factories, the copper-to-board edge clearance requirement is 0.2 mm. Failure to meet the factory's requirements could increase the risk of exposed copper on the edge of the boards or damaged traces/pads, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. Copper-to-board edge spacing of 7.89mil was detected in your design. It is recommended to increase the spacing to at least 0.25 mm for edge routing and 0.4 mm for v-cuts (v-cut spacing may depends on board thickness).		104.83,-136.96	28	Risk
8	Drill Diameter_Slot Aspect Ratio	1.5,2,2	0.20 mm	Slots with aspect ratio of 1.89 were detected in your design. This could increase the risk of incomplete drilling of the slot, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. The ratio should be increased to at least 2:1		176.72,-135.09	6	Warning

9	Drill Diameter_Smallest Drill Size	0.19,0.25,0.3	0.20 mm	Min.Drill:0.20mm in diameter were detected in your design. Smaller drill bits necessitate more frequent replacements, increasing the likelihood of missed holes, misalignment, and rough hole walls. This diminishes manufacturing efficiency, reduces yield, and impacts board reliability. Holes with diameters of 0.1mm or less require laser drilling and must adhere to stringent board width requirements. It is advisable to increase the diameter to at least 0.2mm or 0.3mm to avoid additional costs		150.89,-87.07	13	Warning
10	Drill Spacing_Different Net Via Spacing	0.2,0.3,0.4	0.24 mm	Different net via spacing of 11.79mil were detected in your design. This could increase the risk of short circuits caused by Conductive Anodic Filament (CAF) electrochemical migration over the lifetime of the PCBs, which decrease manufacturing efficiency and yield, and affect the reliability of the boards. It is recommended to increase the spacing to at least 12 mil.		123.71,-97.42	57	Warning

11	Drill Spacing_Same Net Via Spacing	0. 19,0.25,0. 3	0.24 mm			138.00,-95.59	402	Warning